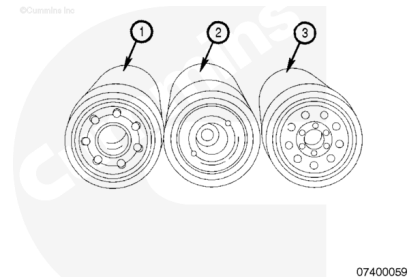


Trainings ▾

Remove

⚠ CAUTION ⚠

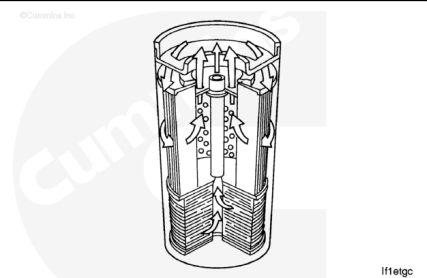
The bypass and combination filters have the same threads.
Verify the correct filter is used for replacement to avoid
damage to the engine.



The external appearance of the full-flow (1), the bypass (2), and the combination (3) filters are the same. The accompanying picture identifies the differences among the three filters.

The full-flow, bypass, and combination filters contain 1 ½ 12-inch threads.

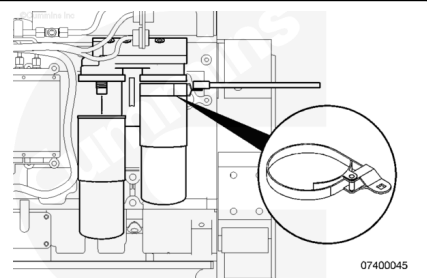
A combination oil filter is used on most engines. The upper section of the filter contains the full-flow filter element while the lower section contains the bypass element.



Note : The following illustrations show the combination oil filter. Use the same procedure when changing the remote bypass oil filters.

Use an oil filter wrench, Part Number 3375049, or equivalent to remove the oil filters.

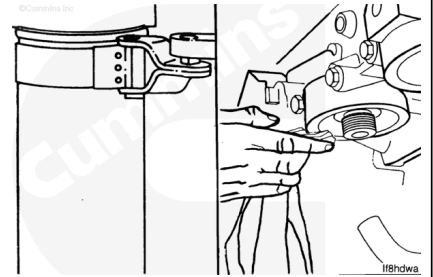
Discard the filters if they are **not** needed for a failure analysis.



Install

Clean the filter head sealing surface with a lint free cloth.

Use clean engine oil to lubricate the gasket surface of the filters.

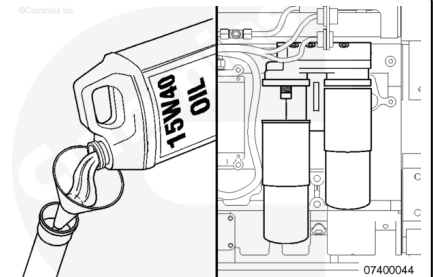


Fill the filters with clean, 15W-40 engine oil.

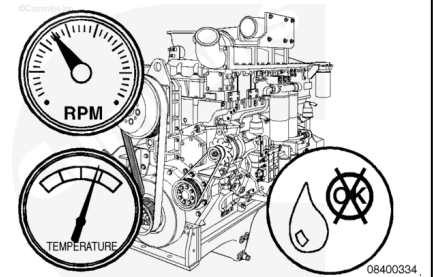
Install the filters on the oil filter head.

Mechanical overtightening can distort the threads or damage the filter element seal.

Turn the filter until the seal contacts the head. Turn an additional $\frac{3}{4}$ -turn to 1 turn.

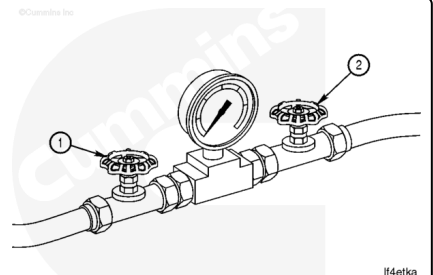


Operate the engine until the oil temperature is at least 65°C [150°F] and check for leaks.



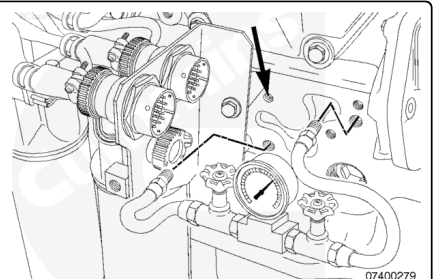
Pressure Differential Test

Use a differential pressure gauge, or one pressure gauge with two oil hoses and two valves to eliminate gauge error. Use a gauge with a minimum pressure capacity of 517 kPa [75 psi].



Install hoses into the before-filter and after-filter oil pressure ports. The ports are located between the oil filter heads and the flywheel housing.

The after-filter oil pressure port is closest to the flywheel housing.



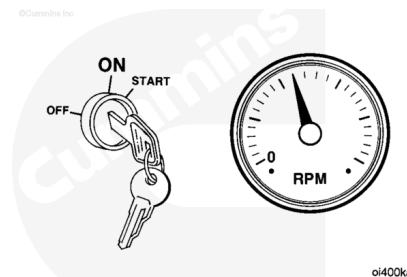
The before-filter oil pressure port is slightly lower and left of the after-filter port.

Note : Applications with air compressors will use the port directly above the before-filter port for after-filter pressure.

Operate the engine at rated rpm and normal operating temperature.

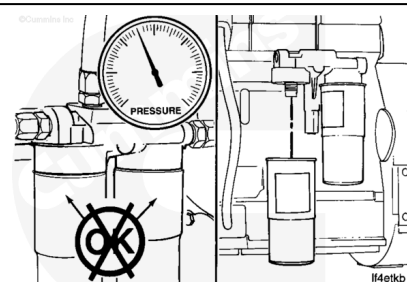
Open the valve to read the pressure after the filter first.

Close the valve and open the other valve to read the oil pressure before the filter.



If the difference in pressure is more than 103 kPa [15 psi], replace the filters.

If the difference in pressure using clean oil filters is more than 69 kPa [10 psi], an oil filter with excessive restriction is being used. Replace the oil filters.



kpa		psi
69	MAX	10